

Simulation Without a Black Box

Outline

Learning Objectives

$M/G/1$ Setup

Pollaczek-Khinchine Formula

$M/D/1$ as Special Case

Higher CV Means More Congestion

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Learning Objectives

By the end of this lecture you will be able to:

- State the M/G/1 model and identify what “G” allows relative to M/M/1.
- Apply the Pollaczek-Khinchine (P-K) formula to compute W_q .
- Derive M/D/1 as a special case and verify it gives half the M/M/1 wait.
- Explain the role of the coefficient of variation (CV) in queue congestion.
- Simulate M/M/1, M/D/1, and M/G/1 (lognormal) at $\rho = 0.8$ and compare.

M/G/1 Setup: General Service Times

M/G/1 assumptions

- **M** — Poisson arrivals, rate λ
- **G** — General service time S with mean $E[S] = 1/\mu$ and variance $\text{Var}[S]$
- **1** — single server, FCFS, infinite buffer
- Stability: $\rho = \lambda/\mu = \lambda E[S] < 1$

The exponential is the only distribution with the memoryless property. With general G , the state space is no longer Markovian — we need the *embedded Markov chain* (departure epochs) to derive steady-state results.

Distribution	$\text{CV} = \sigma/\mu_S$
Deterministic (D)	0
Erlang- k	$1/\sqrt{k}$
Exponential (M)	1
Hyperexponential	> 1
Lognormal (typical)	1.0–3.0

CV quantifies service-time *variability* relative to the mean.

The Pollaczek-Khinchine (P-K) Formula

P-K mean value formula

$$W_q = \frac{\lambda (\text{Var}[S] + (E[S])^2)}{2(1 - \rho)} = \frac{\lambda E[S^2]}{2(1 - \rho)}$$

Equivalently, using $E[S] = 1/\mu$ and $\text{CV} = \sigma_S \mu$:

$$W_q = \frac{\rho}{\mu(1 - \rho)} \cdot \frac{1 + \text{CV}^2}{2} = W_q^{\text{M/M/1}} \cdot \frac{1 + \text{CV}^2}{2}$$

- $\text{CV} = 1$ (exponential): P-K recovers M/M/1 exactly.
- $\text{CV} = 0$ (deterministic): P-K gives $W_q = W_q^{\text{M/M/1}}/2$.
- $\text{CV} > 1$ (heavy-tailed): $W_q > W_q^{\text{M/M/1}}$.

Arrival process variability (λ) and service variability (CV) both drive congestion.

M/D/1: Deterministic Service ($CV = 0$)

With constant service time $S = 1/\mu$ (no randomness): $\text{Var}[S] = 0$, $E[S^2] = (1/\mu)^2$.

$$W_q^{M/D/1} = \frac{\lambda(1/\mu)^2}{2(1-\rho)} = \frac{\rho/\mu}{2(1-\rho)} = \frac{1}{2} W_q^{M/M/1}$$

Interpretation

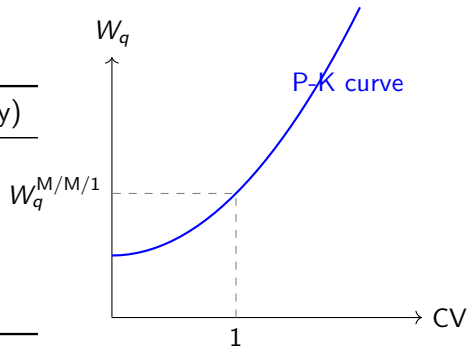
Eliminating service-time randomness *halves* the queue wait — at no additional cost (same mean throughput). This is the most cost-effective queue improvement available without adding servers.

- In practice: standardise procedures, use fixed-duration slots (appointments), automated tellers.
- Reducing CV from 1 to 0.5 achieves 37.5% reduction in W_q (at $\rho = 0.8$).

Higher CV $\Rightarrow$ More Congestion

At $\rho = 0.8$, $\mu = 1$ ($W_q^{M/M/1} = 4.0$):

Distribution	CV	W_q (theory)
M/D/1 (constant)	0	2.00
M/E2/1 (Erlang-2)	0.71	3.00
M/M/1 (exponential)	1.0	4.00
M/G/1 (lognormal, $\sigma = 1$)	1.31	≈ 5.62
M/G/1 (Pareto, heavy tail)	2.5	≈ 14.50



Simulation: M/M/1 vs. M/D/1 vs. M/G/1 at $\rho = 0.8$

```
import numpy as np, simpy

def make_svc_fn(dist, rng, mu=1.0):
    if dist == "MM1":
        return lambda: rng.exponential(1/mu)
    elif dist == "MD1":
        return lambda: 1/mu          # constant
    elif dist == "MG1_lognorm":
        # lognormal with mean=1/mu, CV approx 1.31
        sigma = 1.0
        m = np.log(1/mu) - sigma**2/2
        return lambda: rng.lognormal(mean=m, sigma=sigma)

results = {}
for dist in ["MM1", "MD1", "MG1_lognorm"]:
    rng = np.random.default_rng(0)
    svc = make_svc_fn(dist, rng)
    # ... standard source/customer SimPy pattern ...
    results[dist] = np.mean(waits)
```

Expected outputs: MM1 $\approx$ 4.0, MD1 $\approx$ 2.0, MG1-lognorm $\approx$ 5.6.

Practical Implications: Variability vs. Capacity

Two levers to reduce W_q

1. **Add capacity:** reduce ρ (add servers or speed up service).
2. **Reduce variability:** lower CV (standardise procedures, use appointments, batch setups).

Lever 2 is often cheaper and faster to implement. Both are captured in P-K:

$$W_q \propto (1 + \text{CV}^2)$$

When simulation beats theory

The P-K formula applies only to M/G/1. For G/G/1 (general inter-arrivals too), use the Kingman approximation or — better — simulation.

$$W_q^{G/G/1} \approx W_q^{M/M/1} \cdot \frac{c_a^2 + c_s^2}{2}$$

where c_a, c_s are CVs of inter-arrivals and service.

Key Takeaways

Core ideas from L13

1. The **P-K formula** $W_q = \lambda E[S^2]/(2(1 - \rho))$ applies to any M/G/1 queue.
2. $W_q^{M/G/1} = W_q^{M/M/1} \cdot (1 + CV^2)/2$ — CV is the key multiplier.
3. M/D/1 (CV=0) achieves **half the M/M/1 wait** at the same load, for free.
4. Heavy-tailed service (CV > 1) dramatically inflates W_q — a common real-world surprise.
5. **Reducing variability is as powerful as adding capacity** — and often cheaper.

Next: L14 — Finite-capacity queues: blocking, truncation, and effective throughput.